

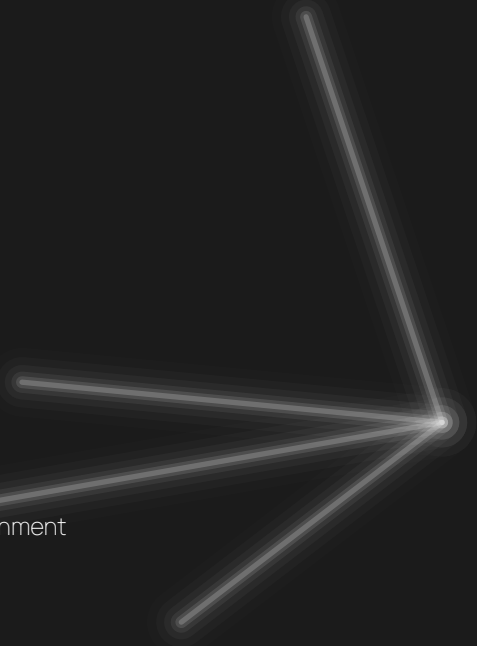
Turning Churn Risk into Revenue

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A Proactive Retention Engine for Company A

GCI World 2026 (Matsuo-Iwasawa Lab, UTokyo) — Final Assignment

Proof of Concept — Data-Driven Business Proposal



Outline

1 Market Context

2 Data & Evidence

3 Modelling

4 Business Proposal

In a mature market, growth comes from keeping customers

- Mobile markets are mature: penetration exceeds 100% (multiple SIMs), so growth comes from ARPU and retention, not new subscribers. [1]
- Acquiring a subscriber costs roughly 5× more than keeping one. [2]
- A 5% lift in retention can raise profit by 25–95%. [3]

The lever

Company A has 100,000 customers and an \$58.72 average monthly bill, but no in-house ML. Even a small drop in churn protects millions in recurring revenue.

Company A snapshot

Customers	100,000
Avg. monthly bill	\$58.72
Avg. tenure	18.8 mo
Avg. handset age	392 days
Churn-risk signals	≈50%

Our task: identify the business problem from the data and propose a solution the client can act on.



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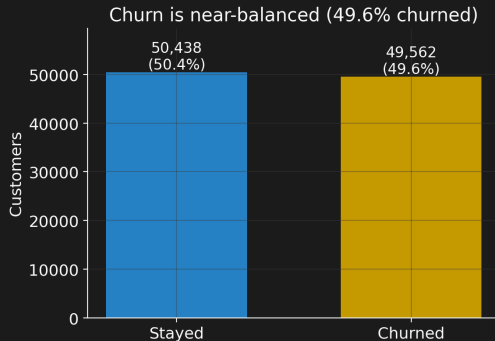
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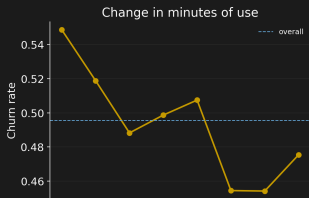
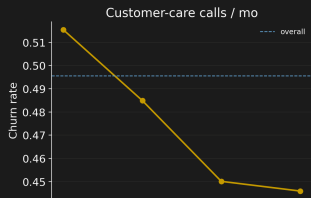
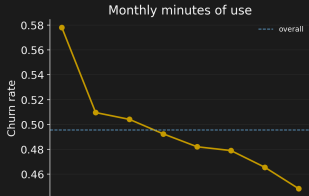
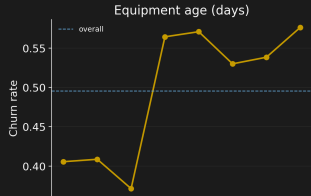
The data, and why churn is the right target

- Two linked tables, joined 1:1 on `Customer_ID`:
 - `Client.csv` — 100k × 50: tenure, plan, handset, demographics
 - `Record.csv` — 100k × 51: monthly usage, billing, call quality, `churn`
- Target = churn (binary). It's the outcome the business most wants to change, and the one tied to revenue.
- Classes are near-balanced (49.6% churn) — so both accuracy and AUC are meaningful here.



What the data says drives churn

Churn rate by feature octile (low → high)



Churn rate rises clearly with:

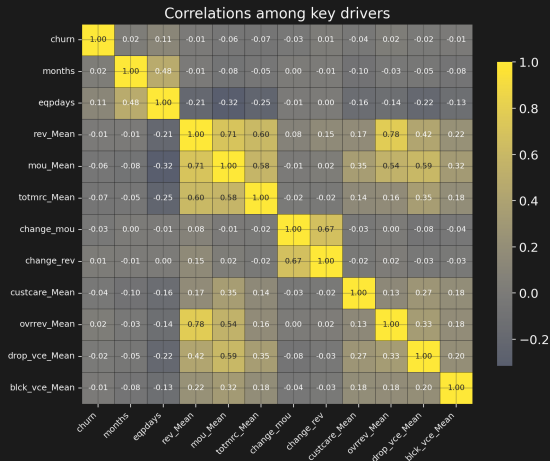
- ageing handsets (equipment age),
- declining usage (change in minutes),

and falls with engagement (customer-care contact, higher usage).

Why it matters

Each of these is something the business can act on — unlike age or location.

No single variable explains churn, so we need a model



- Correlations with churn are individually weak ($|r| < 0.1$); revenue/usage columns are near-duplicates of each other.
- Signal is diffuse and non-linear: simple rules (like “flag old phones”) only catch part of it.

Implication

A multivariate model that combines weak signals will rank churn risk far better than any single threshold.

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From business question to ML task

Task. Binary classification; output a per-customer churn-probability score the business can rank and target.

Feature engineering.

- Dropped **Customer_ID** (no predictive value).
- Encoded 21 categorical columns.
- Built 11 interpretable features — ratios, shares and momentum: e.g.
`usage_momentum`, `rev_per_min`,
`overage_share`, `eqp_per_tenure`,
`drops_per_call`.

Models compared.

- Logistic Regression — transparent linear baseline (median-imputed, scaled).
- XGBoost, CatBoost, HistGradientBoosting — gradient-boosted trees; handle missing values and non-linear interactions natively.
- Rank-blend ensemble of the three — the final model.

Metrics

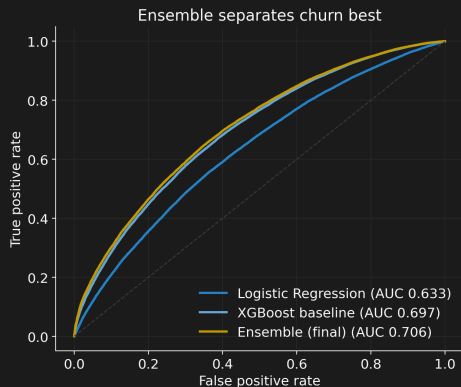
ROC-AUC (ranking quality — what targeting needs) plus accuracy (meaningful because classes are balanced), both via 5-fold stratified cross-validation (honest out-of-fold predictions).

Results: engineered features and the ensemble both lift AUC

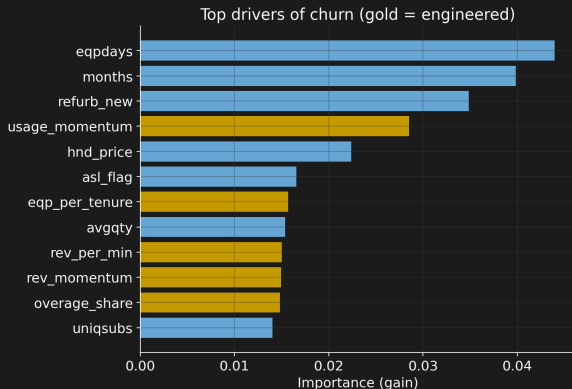
Model	Acc.	ROC-AUC	F1
Logistic Regression	0.596	0.633	0.587
XGBoost (no features)	0.641	0.697	0.642
XGBoost + features	0.648	0.706	0.645
CatBoost + features	0.648	0.704	0.646
Ensemble (final)	0.648	0.706	0.647

Chosen model

Ensemble (CatBoost + XGBoost + HistGB) – ROC-AUC 0.706, Accuracy 0.648, by 5-fold cross-validation over all 100,000 customers (vs. 0.50 for a random baseline). Feature engineering alone lifted AUC from 0.697 to 0.706.



The model points to levers the business can pull



Top driver	Business action
Equipment age, refurb, handset price	Proactive handset upgrade offers
Usage & revenue momentum	Win-back on usage decline
Overage share, rev/min	Right-plan nudges

Gold bars are engineered features. Several reach the top tier (usage momentum is the #4 driver), so the feature work paid off. Tenure and account type rank high too, but as risk markers, not levers.

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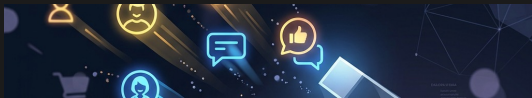
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Proposal: a Proactive Retention Engine

How it works. Score the base monthly; route the highest-risk customers to the matched offer (upgrade / win-back / re-plan).

Why targeting pays. Most of the churn sits in a small slice:

- Top 20% by score $\Rightarrow$ 74% churn density (1.5 $\times$ lift), capturing 30% of all churners.
- Retention spend stops subsidising the half who would stay anyway.



Year-1 impact (stated assumptions)

Target (top 20%)	20,000
At-risk within target	14,880
Save rate (assumed 12%)	1,786 saved
Revenue retained @ \$705/yr	\$1.26M
Program cost @ \$40/offer	\$0.80M
Net Year-1 benefit	$\approx$ \$0.46M
ROI	$\approx$ 57%

Conservative: industry save rates run 10–20%; benefit compounds in later years.

Roadmap and risks

From PoC to production

- 1 Pilot one region with an A/B holdout to measure true uplift.
- 2 Integrate monthly scoring into the CRM / offer engine.
- 3 Scale nationally; refresh the model quarterly.



Risks & mitigations

- Model drift → scheduled retraining, monitoring.
- Wasted discounts → move to uplift modelling (target persuadables, not sure-leavers).
- Privacy → consent-based use, no sensitive attributes as levers.

Bottom line

On its own, the score changes nothing. The value is the targeted retention program it drives: 100k rows become a ranked action list with a defensible, positive ROI.

References

- 1 GSMA. The Mobile Economy 2024. GSMA Intelligence. (Market maturity & ARPU-led growth.)
- 2 Reichheld, F. F. The Loyalty Effect. Harvard Business School Press, 2001. (Acquisition vs. retention cost.)
- 3 Reichheld, F. F. & Sasser, W. E. (1990). "Zero Defections: Quality Comes to Services." Harvard Business Review 68(5), 105–111.
- 4 Chen, T. & Guestrin, C. (2016). "XGBoost: A Scalable Tree Boosting System." Proc. KDD '16, 785–794.
- 5 Prokhorenkova, L. et al. (2018). "CatBoost: unbiased boosting with categorical features." NeurIPS 31.
- 6 Pedregosa et al. (2011). "Scikit-learn: Machine Learning in Python." JMLR 12, 2825–2830.
- 7 Company A telecom dataset, GCI World 2026 Final Assignment (Matsuo-Iwasawa Lab, UTokyo).

Appendix A: Data preparation, cleaning, and target choice

Source. Two files joined 1:1 on **Customer_ID** into a single 100,000 × ~100 table. **Client.csv** carries account-level fields (length-of-relationship **lor**, equipment age **eqpdays**, handset price **hnd_price**, demographics, plan); **Record.csv** carries mean monthly usage, billing and network-quality metrics plus the **churn** flag and **months**.

Why churn is the target. The flag is unusually balanced (~50/50), which keeps accuracy interpretable, and it maps directly to the retention problem the business actually faces. Alternative targets (e.g. ARPU regression) were considered but answer a different question and would not produce a rankable at-risk list.

Identifier. **Customer_ID** is a unique key with no predictive content and is dropped before any modelling.

Redundancy. The ~100 columns include several near-duplicate revenue and usage variants. Highly correlated columns were pruned to cut multicollinearity and noise without losing signal.

Missing values. Columns that were heavily missing were dropped rather than imputed, since imputing a mostly-empty field invents data. The gradient-boosted trees handle remaining gaps natively; the logistic baseline alone used median imputation and standardisation.

Leakage guard. Any field that could only be known after a churn event was excluded. Tenure fields (**months**, **lor**) are retained because they are known up front, but they are treated as risk markers, not levers — the business cannot act on them.

Key assumption. The monthly means in **Record.csv** represent a stable, comparable observation window across customers.

Appendix B: Feature engineering and modelling detail

Engineered features (11). Ratios, shares and momentum built to be interpretable to the business:

- **usage_momentum** — recent vs. baseline usage trend; a sustained decline is an early churn signal.
- **rev_per_min** — revenue per minute, a proxy for plan fit.
- **overage_share** — overage charges as a share of the bill; flags customers on the wrong plan.
- **eqp_per_tenure** — equipment age relative to tenure; isolates ageing-handset risk.
- **drops_per_call** — dropped-call rate, a network-quality proxy.

plus further ratio/share terms in the same family.

Categoricals. 21 categorical columns encoded — LabelEncoder as the baseline, with CatBoost additionally using native categorical handling.

Models. A transparent Logistic Regression baseline; XGBoost, CatBoost and HistGradientBoosting as non-linear learners; and a rank-blend ensemble of the three (averaging rank-normalised probabilities) as the final model, chosen for lower variance rather than a higher headline score.

Validation. 5-fold stratified cross-validation over all 100,000 rows; metrics are computed on honest out-of-fold predictions. Fold-safety is enforced: any group-aggregate or target-derived transform is fit on the fold-training slice only, then applied to the held-out fold and test — so no validation information leaks into training.

Metric choice. ROC-AUC is primary because targeting needs good ranking, not a hard cut-off; accuracy and F1 are reported as secondary checks and are meaningful only because the classes are balanced.

Outcome. Feature engineering lifted AUC from 0.697 to 0.706, and the ensemble matched the best single model (0.706) with more stable fold-to-fold behaviour.

Appendix C: Business-case assumptions and ROI derivation

Targeting logic. The whole base is scored monthly; action is taken only on the top 20% (20,000 of 100,000). That quintile carries a 74% churn density — a 1.5x lift over the ~50% base rate — and captures roughly 30% of all churners. Concentrating spend here stops the program subsidising the half of customers who would stay anyway.

Year-1 arithmetic.

- At-risk within target: $20,000 \times 74\% = 14,880$.
- Save rate 12% (assumed; industry runs 10–20%) $\Rightarrow 1,786$ customers saved.
- Revenue retained: $1,786 \times \$705/\text{yr} \approx \1.26M .
- Program cost: $20,000 \times \$40/\text{offer} = \0.80M .
- Net Year-1 $\approx \$0.46\text{M}$; ROI $\approx 57\%$.

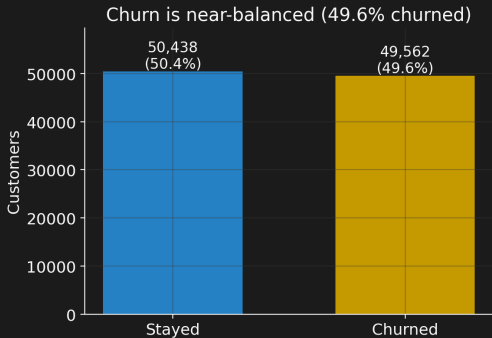
Where the inputs come from.

- \$705/yr is the annualised mean revenue per retained customer, derived from the dataset's billing fields.
- \$40/offer is a blended cost across the three offer types (handset upgrade / win-back / re-plan), charged against the full targeted quintile, not just savers — a deliberately conservative treatment.
- The 12% save rate sits at the low end of the cited industry range.

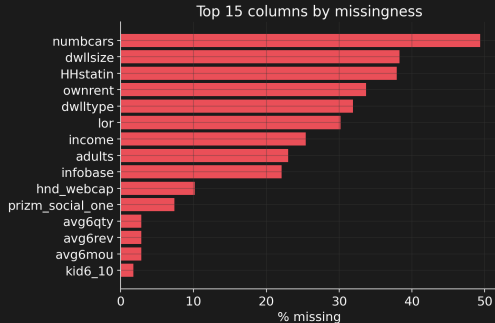
Sensitivity. At a 10% save rate the program is near break-even; at 20% the net benefit roughly doubles. The Year-1 figure also excludes downstream lifetime value — saved customers persist into later years, so the steady-state return is higher.

Caveats. This is a cross-sectional snapshot; it establishes correlation, not proven causal uplift. The roadmap therefore starts with a regional pilot carrying an A/B holdout, which measures the true incremental save rate before any national scale-up.

Appendix D: EDA, target balance and data quality

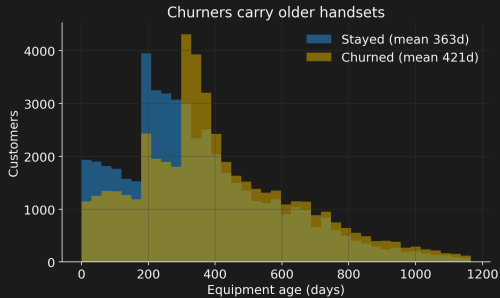


Target is near-balanced (49.6% churned) — accuracy is interpretable and no resampling is needed, a key reason churn is a sound target.

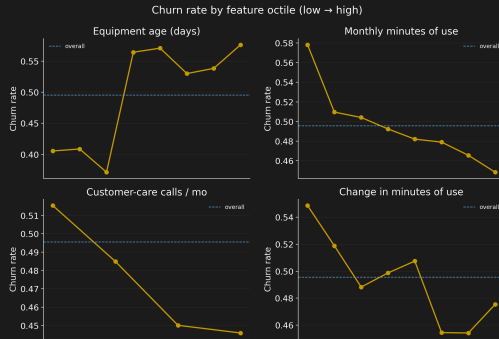


A handful of columns are heavily empty; these are dropped rather than imputed, since imputing a mostly-empty field invents data.

Appendix E: EDA, univariate churn signals

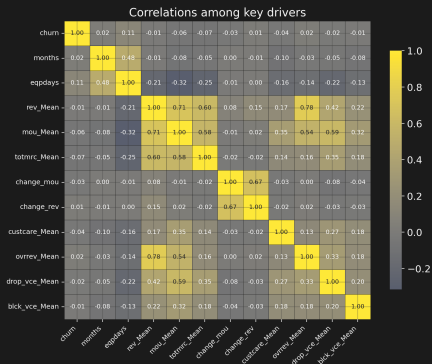


Churners carry older handsets — equipment age is an actionable retention lever (proactive upgrade offers), unlike tenure.

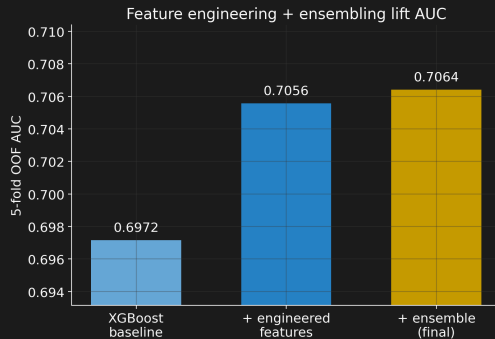


Churn rate by octile of four key drivers: each moves monotonically, but no single variable separates churn — motivating a multivariate model.

Appendix F: EDA correlations and the modelling AUC lift

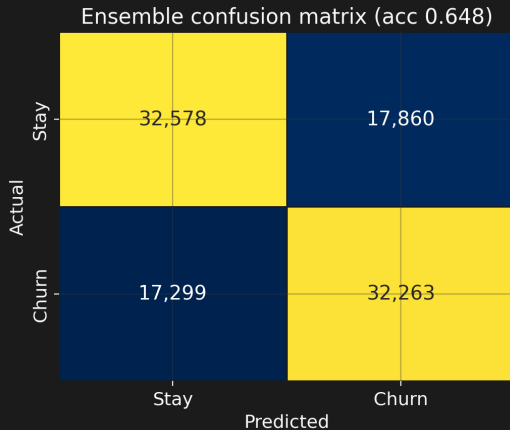


Drivers correlate only weakly with churn and partly with each other — supporting both the multivariate model and the pruning of near-duplicate columns.



5-fold out-of-fold AUC: engineered features (0.697→0.706) then the rank-blend ensemble each add lift.

Appendix G: Model diagnostics, ensemble confusion matrix



Final rank-blend ensemble at a 0.5 threshold (accuracy 0.648). Errors are roughly balanced across stay / churn, consistent with the balanced target — and for targeting we rank by score rather than apply this hard cut-off.